

Program: Diploma in Artificial Intelligence & Machine Learning	
Course Code: 5348	Course Title: Big Data Analytics Lab
Semester: 5	Credits: 1.5
Course Category: Programme Core Course	
Periods per week:: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Optimize decisions and improve efficacy in business administration with Big data analytics
- Impart the architectural concepts of Hadoop and introduce map-reduce paradigm
- Familiarize programming tools PIG and HIVE in the Hadoop ecosystem.
- Implement best practices for Hadoop development in real-world problems.

Course Prerequisites:

Topic	Course code	Course name	Semester
Sound knowledge in Python Programming	1008	Introduction to IT Systems Lab	I
Sound knowledge in Python Programming	3347	Python Programming Lab	III
Basic Concepts of Database Management Systems	3345	Data Structure Lab	III

Course Outcomes:

On completion of the course, the student will be able to:

CO_n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the installation of VMWare & Hadoop operating modes.	8	Applying
CO2	Practice the basic commands of LINUX Operating System & file management in Hadoop.	8	Applying
CO3	Illustrate Map Reduce Paradigm & its programming applications in solving real-time problems.	12	Applying
CO4	Perform the installation and applications of programming tools like PIG & HIVE in the real-world Hadoop Ecosystem.	14	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	2	2					
CO3	3	3					
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the installation of VMWare & Hadoop operating modes.		
M1.01	Demonstrate the installation of VMWare to setup the Hadoop environment and its ecosystems	2	Applying

M1.02	Demonstrate setting up and Installing Hadoop in its three operating modes. i. Standalone. ii. Pseudo distributed. iii Fully distributed.	4	Applying
M1.03	Use web-based tools to monitor your Hadoop setup.	2	Applying
CO2	Practice the basic commands of LINUX Operating System & file management in Hadoop.		
M2.01	Practice the basic commands of LINUX Operating System – File/Directory creation, deletion, update operations- use commands like pwd,mkdir,rmdir,ls,cd,touch,cat,rm,cp,mv etc	4	Applying
M2.02	Implement the following file management tasks in Hadoop: i. Adding files and directories ii. Retrieving files iii. Deleting files	4	Applying
	Lab Exam – I	1.5	
CO3	Illustrate Map Reduce Paradigm & its programming applications in solving real time problems.		
M3.01	Practice basic word count Map Reduce program to understand Map Reduce Paradigm.	2	Applying
M3.02	Write a Map Reduce program that mines weather data.	3	Applying
M3.03	Practice matrix multiplication with Hadoop Map Reduce.	3	Applying
M3.04	Apply K-means clustering using Map Reduce	4	Applying
CO4	Perform the installation and applications of programming tools like PIG & HIVE in the real world Hadoop Ecosystem		
M4.01	Illustrate the installation of PIG	2	Applying

M4.02	Write Pig Latin scripts sort, group, join, project, and filter your data. Demonstrate how to a. Run the Pig Latin Scripts to find Word Count b. Run the Pig Latin Scripts to find a max temp for each and every year.	3	Applying
M4.03	Illustrate the installation of HIVE	2	Applying
M4.04	Use Hive to create, alter, and drop databases, tables, views, functions, and indexes.	2	Applying
M4.05	Open Ended Experiments**	5	Applying
	Lab Exam -2	1.5	

Text / Reference

T/R	Book Title/Author
T1	Jay Liebowitz, —Big Data and Business Analytics Laboratory, CRC Press.
T2	Seema Acharya, Subhashini Chellappan, "Big Data Analytics", Wiley, 2015
R1	Tom White, — Hadoop: The Definitive Guide Third Edition, O'Reilly Media, 2012

Online Resources

Sl.No	Website Link
1	https://www.vmware.com/tryvmware/?p=workstation-w
2	https://www.tutorialspoint.com/apache_pig/index.htm
	https://www.tutorialspoint.com/hive/index.htm
3	http:// apache.mirrors.lucidnetworks.net/hadoop/
3	https://hadoop.apache.org/docs/r1.2.1/mapred_tutorial.html